

LAN-Based Blind SSRF Attack Primitive for Windows Systems (switcheroo)

LAN-Based Blind SSRF Attack Primitive for Windows Systems (switcheroo) - initstring, initblog.com.

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Content

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Introduction

Overview

Unauthenticated attackers on a local network can force stock Windows systems to perform arbitrary HTTP GET requests, including to the target's localhost interface. No user interaction is required. No IIS installation is required. Network Discovery must be enabled to trigger the exploit (usually on by default for private networks). The response cannot be viewed by the attacker, making this a "Blind Server-Side Request Forgery" vulnerability.

This provides an exploit primitive that can be used for attacking services like REST APIs. Applications deployed only to localhost, or those on protected network segments, are exposed to attackers due to this OS vulnerability.

This is done by responding to SSDP multicast discovery requests, which Windows systems send out automatically and at regular intervals. These requests ask for a URL to learn more about available shared devices on a LAN. Windows will ignore the host portion of any advertised URL, connecting only to the host who provided the advertisement. However, a malicious host can reply to that connection with an HTTP 301 redirect. Windows will process the redirection completely, accepting any arbitrary host (including 127.0.0.1) and URL path.

[\[image: diagram\]](#)

While this write-up is specific to the vulnerable SSDP service in Windows, it is likely that many other applications using their own SSDP stack are vulnerable as well.

By itself, this is not exactly an earth-shattering vulnerability. In fact, Microsoft has responded saying that they do not intend to fix it (see full disclosure info at the end of this blog). I thought it was interesting enough to share, and hope that others may find some functionality in the primitive itself.

Exploit POC code is available in the GitLab repository [here](#).

Vulnerable Windows OS Versions

The exploit POC was tested on the following versions of Windows, all of which were vulnerable:

- Windows 10 Enterprise Insider Preview v1903 (build 18932.1000)
- Windows 10 Enterprise v1903 (build 18362.30)
- Windows Server 2016 Standard v1607 (build 14393.693)

Details

SSDP Discovery Process

Simple Service Discovery Protocol (SSDP) is used by Operating Systems (Windows, MacOS, Linux, IOS, Android, etc) and applications (Spotify, Youtube, etc) to discover shared devices on a local network. It is the foundation for discovering and advertising Universal Plug & Play (UPNP) devices.

The cool thing about SSDP, for an attacker, is that there are no real security mechanisms built in. A machine using SSDP to discover devices will send clear-text messages to everyone on the LAN and trust any device that replies.

Devices attempting to discover shared network resources will send a UDP multicast out to 239.255.255.250 on port 1900. The source port is randomized. An example request looks like this:

```
M-SEARCH * HTTP/1.1
Host: 239.255.255.250:1900
ST: upnp:rootdevice
Man: "ssdp:discover"
MX: 3
```

Any device on the local area network can respond to that request, providing a URL that points to an XML document called a ‘Device Descriptor’. This reply looks like the following:

```
HTTP/1.1 200 OK
CACHE-CONTROL: max-age=1800
DATE: Tue, 16 Oct 2018 20:17:12 GMT
EXT:
LOCATION: http://192.168.1.214:8888/ssdp/device-desc.xml
OPT: "http://schemas.upnp.org/upnp/1/0/"; ns=01
01-NLS: uuid:7f7cc7e1-b631-86f0-ebb2-3f4504b58f5c
SERVER: UPnP/1.0
ST: upnp:rootdevice
USN: uuid:7f7cc7e1-b631-86f0-ebb2-3f4504b58f5c::upnp:rootdevice
BOOTID.UPNP.ORG: 0
CONFIGID.UPNP.ORG: 1
```

Once the Windows system receives this reply, it will then initiate an HTTP GET request to the location specified in the `LOCATION` header above.

Abusing SSDP for SSRF

While the example above provides an IP address (192.168.1.214), Windows will completely ignore that. For example, if the `LOCATION` header specified `http://127.0.0.1/device.xml`, Windows would pay attention only to the `/device.xml` and perform an HTTP GET to that URL on the remote host that responded to the discovery request. This is a way to prevent SSRF.

However, if that remote host were to respond with an HTTP 301 redirect when queried at `/device.xml`, Windows will trust the full URL provided in a new `LOCATION` header.

This can be demonstrated as follows:

- Connect the attacking box, running Linux, to the same LAN subnet as a Windows machine.
- On the Windows machine, run some listener to verify SSRF, like `ncat` as follows:

```
ncat.exe -nlvp 4444
```

- Run the following from the Linux attack box, replacing eth0 with the appropriate LAN interface:

```
python3 switcheroo.py -i eth0 -u http://localhost:4444/ssrf_pwn -t ""
```

- SSDP Discovery happens on a regular basis in Windows. So you can simply wait, or do the following on the Windows target to speed things up:
- Open Windows Explorer
- Select “Network” from the left
- Click the refresh button in the address bar

From here, you will see the Windows machine connect to itself on localhost, which should be protected. This attack can be customized with switcheroo.py by providing an SSRF URL targeted at any application.

[\[image: screenshot\]](#)

Weaponization

My free time is a bit limited right now due to a new gig and just life in general. I’m releasing this info and POC in the hopes that other hackers can build some cool exploit chains with it. I would love to hear feedback from folks who have success.

The GET is performed as LOCAL SERVICE, if that’s helpful to you Windows researchers out there. Thanks to Elad Shamir for answering my questions on this.

More Info

This discovery came as a byproduct of my earlier research into SSDP vulnerabilities, which you can read more about on the evil-ssdp page [here](#).

Disclosure Timeline

- Initial report to Microsoft: 5th August 2019
- Microsoft confirms receipt of report: 7th August 2019
- Microsoft stated this is not a vulnerability: 9th August 2019

This was my first disclosure to Microsoft. I had hoped it would fit into the “security feature bypass” category. SSDP has a security feature that ignores the host address on discovery replies. The bypass, in my opinion, is that filter is not applied on a 301 redirect. Yes, I know this isn’t exactly the next BlueKeep, but it still seems a bit risky to allow anyone on the LAN to trigger communications with a loopback interface.

Oh well. :)

[\[image: screenshot\]](#)

